Transformer Models and Ethical Challenges in AI

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Case Study : Ethical Challenges in AI-Powered Loan Approval System

Future Trust Bank, a leading private sector bank, introduced an AI-powered loan approval system in 2025 to improve speed, reduce paperwork, and automate decision-making. Earlier, loan officers manually reviewed customer applications based on salary, repayment history, credit score, job stability, and supporting documents. The manual process usually required 5–7 working days.

To improve efficiency, the bank implemented an AI model designed to approve or reject loan applications within 10 minutes using customer data such as age, income, city, occupation, transaction history, loan history, and credit score.

Problem Statement:

- Within three months of implementation, several customer complaints were received:
- Female applicants with similar financial profiles as male applicants were rejected more frequently.
- Applicants from Tier-2 and Tier-3 cities had lower approval rates than metro city applicants.
- Some customers with poor repayment history were unexpectedly approved.
- Rejected applicants received no meaningful explanation except “system decision.”
- Social media criticism increased regarding unfair automated decisions.

Internal Audit Findings:

The internal review team identified the following issues:

- Historical data used for model training reflected past lending bias.
- The model learned correlations between city, gender, and approvals.

- Explainability tools were not implemented.
- Accuracy was acceptable overall, but fairness across demographic groups was weak.
- No regular ethical audit or fairness monitoring process existed.

Business Impact:

- Reputational damage
  - Decline in customer trust
  - Regulatory scrutiny risk
  - Increased grievance handling cost
  - Delay in digital transformation goals

QUESTION-ANSWERS:

QUES.1 Identify the key ethical issues present in this case.

- Discrimination- Discrimination against female applicants as they faced rejection for similar financial profiles as males.
- Accountability- The system always replied with "System Decisions" and didn't gave a proper reason for the rejection.
- Historical data bias- System gave responses by learning past data which created biasness.

QUES.2 Why did the AI system produce unfair outcomes?

- Past learnt patterns- AI-Systems work by learning past data, so it rejected those applications that seemed similar to earlier rejected applications.

- Lack of Fairness Constraints: No mechanisms were in place to ensure equal treatment across demographic groups.

- Absence of Monitoring: There was no continuous fairness audit to detect and correct bias after deployment.

QUES.3 Is model accuracy alone sufficient to evaluate success? Justify your answer.

Accuracy is necessary but accuracy alone isn't enough for a very critical task like loan approval. For Example- It will be considered accurate by a machine to reject loan to most of the females applicants or of the people living in Tier-1 and Tier-2 cities by studying the past data, but legally and ethically it isn't appropriate.

QUES.4 How can explainability improve customer trust?

Explainability can surely improve customer's trust because if applicants are just provided with the statement of 'System decision' they would be really unhappy and it will affect Bank's reputation. Instead applicants should be clearly told about the reasons for which their application has been rejected. Also, if there is a possibility, they should be informed what could be done to correct the same.

QUES.5 Who should be accountable for wrong AI decisions?

The Primary accountability lies in the hands of the Management of the Bank as in the eyes of law they are responsible for their system.

Then comes the Data Governance team who is responsible for feeding the data into the system and didn't train it accordingly.

QUES.6 Suggest fairness metrics suitable for this case.

To evaluate and improve fairness, the following metrics can be used:

- Demographic Parity: Approval rates should be similar across all different groups For example-gender, city.
- Equal Opportunity: Applicants with similar financial profiles should have equal chances of approval.
- Disparate Impact Ratio: Measures whether one group is disproportionately affected.

- False Positive/Negative Rates: Ensures error rates are consistent across groups.

These metrics help identify and reduce bias in AI decision-making.

QUES.7 Recommend practical steps to improve the system.

Future Trust Bank can take several steps to improve the system:

- Data Cleaning: Remove or reduce bias in training data.
- Feature Selection Review: Avoid using sensitive attributes like gender or irrelevant attributes.
- Implement Explainable AI Tools: Use techniques such as SHAP or LIME for better transparency.
- Regular Ethical Audits: Continuously monitor fairness and performance.
- Human interference Approach: Include human review for borderline or high-risk decisions.
- Bias Testing Before Deployment: Evaluate by testing models for fairness before implementation.

QUES.8 As a manager, would you fully automate loan approvals? Explain your decision.

As a manager, I would not recommend fully automating loan approvals.

While AI can significantly improve efficiency and reduce processing time, complete automation can lead to ethical risks, bias, and loss of human judgment. Financial decisions often involve complex, contextual factors that AI may not fully understand.

A hybrid approach is more suitable:

- AI can handle initial screening and data analysis.
- Human experts should review critical or borderline cases.

This approach ensures a balance between efficiency and ethical responsibility, reducing risks while maintaining trust.